Analysis and Geometry on Manifolds

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1 Motivation

1.1 Symmetric Spaces in the Wild

Élie Cartan published the first in a series of 2 papers titled "Sur une classe remarquable despaces de Riemann". A century later, these spaces now known as Riemannain symmetric spaces continued to intriegue, delight and mystify geometers, Lie/group theorist, dynamist, and number theorists.

Definition 1.1. Let (M, g) be a connected Riemannian manifold of dimension positive. M is said to be a (Riemannian) locally symmetric space if for every point $o \in M$, there is an open neighborhood $U_o \subseteq M$ of o and an isometry $s_o : U_o \to U_o$ such that o is the unique fixed point of s_o in U_o and $s_o^2 = \mathrm{id}_{U_o}$, (i.e. it is an involution).

M is a (Riemannian) (globally) symmetric space if for any point $o \in M$, the involution s_o above extends to an isometry of the whole space M. We omit globally if it is clear from the context and simply say symmetric.

Remark 1.1. s_o being locally an involution in a chart forces the extension to also be an involution. However, s_o might admit other fixed point away from o.

It will turn out that the universal cover of a complete Riemannian locally symmetric space is automatically globally symmetric.

In particular every complete locally symmetric spaces can be obtained as a quotient of a globally symmetric space by a covering space action (i.e. discrete actions by isometries where no non-trivial group element has a fixed point).

Example 1.1. The complete simply connected Riemannai manifold of constant sectional curvature are globally symmetric. These are upto scaling, Euclidean spaces $\mathbb{E}^n$, spheres $\mathbb{S}^n$, or hyperbolic spaces $\mathbb{H}^n$.

We will see that symmetric spaces are homogeneous meaning the isometry groups $\mathrm{Isom}(M)$ of M act transitively on M. That is for any $p, q \in M$, there is an isometry $f : M \to M$ such that $f(p) = q$.

This course will focus on simply connected symmetric spaces of everywhere non-positive sectional curvature. These spaces are non-compact with the fundamental group of these compact quotients (i.e. the closed non-positiely curved locally symmetric spaces). Any natural examples of finitely generated infinite groups.

Among the non-positively curved symmetric spaces, those of non-compact type moreover admit finite volue quotinets that are not compact.

Among the simply connected Riemannian manifold of non-positive curvature, those that are globally symmetric indeed factorize by the properties that they are homogeneous and admit compact quotient.

Theorem. Let M be a closed non-positively curved Riemannian manifold. If the universal cover $\widetilde{M}$ is homogeneous then $\widetilde{M}$ is globally symmetric. Hence M is locally symmetric.

There are indeed not-so-many procedures for constructing closed non-positively curved Riemannian manifold that are locally symmetric.

Remark 1.2 (Open Problem)**.** Is there a natural number k such that for any closed non-positively curved Riemannian manifold M, the fundamental group of M contains a subgroup isomorphic to the fundamental group of a closed non-positively curved locally symmetric space of dimension at least $\dim M - k$.

1.2 Remarks on Negatively Curved Locally Symmetric Spaces

Let (M, g) be a closed Riemannian manifold of (strict) negative curvature. We denote,

$$P_g(R) := \left\{ \substack{\text{Primitive oriented closed} \\ \text{geodesics in } (M,g) \text{ of length at most } R} \right\}$$

We further define,

$$h(g) := \lim_{R\to\infty} \frac{1}{R} \log(\#P_g(R)).$$

Let us denote $B_{\widetilde{g}}(x, R)$ to be the ball in the universal cover $\widetilde{M}$ centered around x with radius R. The above quantity can also be described by

$$h(g) = \lim_{R\to\infty} \frac{1}{R} \log(\text{vol}(B_{\widetilde{g}}(x, R))).$$

Theorem 1.1. Let (M, g) be a closed negatively curved Riemannian manifold and suppose M admits a (necessarily negatively curved) locally symmetric metric g_0. Assume that g and g_0 are normalized so that the maximum sectional curvature for each g and g_0 is -1, then $h(g) > h(g_0)$ with equality hold if and only if g is locally symmetric.

Theorem 1.2. Same assumption but this time instead of normalizing to that maximum symmetric curvatures coincide, normalize so that $\text{vol}(M, g) = \text{vol}(M, g_0)$. Then the same conclusion holds.

If (M, g) is closed and negatively curved, there is a sectional geodesic flow-invariant probability measure on T^1M, namely the Liouville measure denoted by μ_L. Any primitive closed geodesic γ in M also gives rise to a geodesic flow-invariant probability measure δ_γ onT^1M, supported and uniformly distributed on the periodic orbit in T^1M projecting to γ.

Theorem 1.3. As $R \to \infty$, the probability measure

$$\frac{1}{\#P_g(R)} \sum_{\gamma\in P_g(R)} \delta_\gamma,$$

convergeto the weak-$\star$-sense to some measure μ_{BM} on T^1M that is geodesic flow-invariant.

Moreover, if g is locally symmetric then $\mu_L = \mu_{\text{BM}}$.

Remark 1.3. It is still unknown that if (M, g) is a closed negatively curved Riemannian manifold for which $\mu_L = \mu_{\mathrm{BM}}$ then g is locally symmetric.

The goal of this course will be to prove the following theorem modulo black boxes, in the case that each of M_i is not of strict negative curvature.

Theorem 1.4. Let $(M_1, g_1), (M_2, g_2)$ be closed non-positively curved locally symmetric manifolds of dimension at least 3. Suppose neither of their universal covers splits as a non-trivial Riemannian product. If their fundamental groups are isomorphic then they are isometric upto scaling.